

## fit\_indexers

This module implements the API to allow users to fit new indexers by wrapping the needed functionality from the string-indexing library.

### DEFAULT\_MAX\_CATEGORIES module-attribute

```
DEFAULT_MAX_CATEGORIES = 10000
```

### DEFAULT\_MIN\_CATEGORY\_SAMPLES module-attribute

```
DEFAULT_MIN_CATEGORY_SAMPLES = 100
```

### logger module-attribute

```
logger = getLogger(__name__)
```

## bulk\_fit

```
bulk_fit(app: App, indexer_name_pattern: str, data: DataFrame, features: List[str], max_categories: int = DEFAULT_MAX_CATEGORIES, min_category_samples: int = DEFAULT_MIN_CATEGORY_SAMPLES) -> List[PulseIndexer]
```

Create several PulseIndexers from a given PySpark DataFrame. This is more performant than calling `fit` directly as operations are submitted as parallel Spark jobs. However, make sure there is enough memory in the executors to hold all indexers in memory.

Indexers with no items are not created on purpose as they would have no use. A warning will be issued for each indexer that was empty.

Parameters:

| Name                                                                                                                  | Type                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Default         |
|-----------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <code>app</code>                   | <code>App</code>         | Pulse Application where the indexer will be created.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <i>required</i> |
| <code>indexer_name_pattern</code>  | <code>str</code>         | A format pattern to generate the indexer name for each indexed feature. For example: "v18_{feature}max_cats". The possible pattern placeholders are: - {feature} : The name of the column being fit - {max_categories} : The actual <code>max_categories</code> parameter value. - {min_category_samples} : The actual <code>min_category_samples</code> parameter value.<br><br>It will show up in the Pulse Lists' pane with that name and the indexer resource prefix ( <code>ds_api.string_indexing.pulse_indexers.PULSE_INDEXER_DESC_PREFIX</code> ). | <i>required</i> |
| <code>data</code>                  | <code>DataFrame</code>   | PySpark DataFrame with data to index. Must have the columns listed in <code>features</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <i>required</i> |
| <code>features</code>              | <code>list of str</code> | Names of the columns to index.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <i>required</i> |

| Name                                                | Type    | Description                                                                                                                                                                                                                                                                                                                                                                                                             | Default                                      |
|-----------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <code>max_categories</code> <a href="#">↗</a>       | (int, ) | <div>default =<br/>ds_api.string_indexing.fit_indexers.DEFAULT_MAX_CATEGORIES</div> <p>Maximum number of categories from the target column to index. Categories with the least number of observations will be filtered out from the indexer mapping until only <code>max_categories</code> are left. Any categories left outside this will be left to the "others" bucket. If set to 0 no such clipping is applied.</p> | <a href="#">DEFAULT_MAX_CATEGORIES</a>       |
| <code>min_category_samples</code> <a href="#">↗</a> | (int, ) | <div>default =<br/>ds_api.string_indexing.fit_indexers.DEFAULT_MAX_CATEGORIES</div> <p>Minimum number of observations that a category must have during fit to be indexed. If set to 0 no such clipping is applied.</p>                                                                                                                                                                                                  | <a href="#">DEFAULT_MIN_CATEGORY_SAMPLES</a> |

Returns:

| Type                              | Description                                  |
|-----------------------------------|----------------------------------------------|
| List of <code>PulseIndexer</code> | The created <code>PulseIndexer</code> Lists. |

## fit [↗](#)

```
fit(app: App, indexer_name: str, data: DataFrame, feature: str, max_categories: int = DEFAULT_MAX_CATEGORIES, min_category_samples: int = DEFAULT_MIN_CATEGORY_SAMPLES) -> PulseIndexer
```

Create a new `PulseIndexer` from a given PySpark `DataFrame`. The Spark `DataFrame` must have a column with that name.

Parameters:

| Name                                          | Type                      | Description                                                                                                                                                                                                                                                                                                                                                                                                             | Default                                |
|-----------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <code>app</code> <a href="#">↗</a>            | <a href="#">App</a>       | Pulse Application where the indexer will be created.                                                                                                                                                                                                                                                                                                                                                                    | <i>required</i>                        |
| <code>indexer_name</code> <a href="#">↗</a>   | str                       | Name to give to the generated <code>PulseIndexer</code> . It will show up in the Pulse Lists' pane with that name and the indexer resource prefix ( <code>ds_api.string_indexing.pulse_indexers.PULSE_INDEXER_DESC_PREFIX</code> ).                                                                                                                                                                                     | <i>required</i>                        |
| <code>data</code> <a href="#">↗</a>           | <a href="#">DataFrame</a> | PySpark <code>DataFrame</code> with data to index. Must have the column <code>colname</code> .                                                                                                                                                                                                                                                                                                                          | <i>required</i>                        |
| <code>feature</code> <a href="#">↗</a>        | str                       | Name of the column to index.                                                                                                                                                                                                                                                                                                                                                                                            | <i>required</i>                        |
| <code>max_categories</code> <a href="#">↗</a> | (int, )                   | <div>default =<br/>ds_api.string_indexing.fit_indexers.DEFAULT_MAX_CATEGORIES</div> <p>Maximum number of categories from the target column to index. Categories with the least number of observations will be filtered out from the indexer mapping until only <code>max_categories</code> are left. Any categories left outside this will be left to the "others" bucket. If set to 0 no such clipping is applied.</p> | <a href="#">DEFAULT_MAX_CATEGORIES</a> |
|                                               | (int, )                   |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                        |

| Name                              | Type | Description                                                                                                                                                                                                                                      | Default                                   |
|-----------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <code>min_category_samples</code> |      | <div><code>default =</code><br/><code>ds_api.string_indexing.fit_indexers.DEFAULT_MAX_CATEGORIES</code></div> <p>Minimum number of observations that a category must have during fit to be indexed. If set to 0 no such clipping is applied.</p> | <code>DEFAULT_MIN_CATEGORY_SAMPLES</code> |

Returns:

| Type                      | Description                    |
|---------------------------|--------------------------------|
| <code>PulseIndexer</code> | The created PulseIndexer List. |

Raises:

| Type                    | Description                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <code>ValueError</code> | If during the indexer fit there are not enough samples to generate any categories at all and the indexer would be empty. |